

Isomorphisms of deterministic $[T, T^{-1}]$ systems

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- 4 "Time" = $\mathbb{Z}$ (no group actions nor flows).

Context

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Consider the Bernoulli shift $\sigma \curvearrowright (\{-1, 1\}^{\mathbb{Z}}, \nu_{\frac{1}{2}, \frac{1}{2}})$. The $[T, T^{-1}]$ system is the transformation

$$(y, x) \mapsto (\sigma(y), T^{y(0)}(x))$$

acting on $(\{-1, 1\}^{\mathbb{Z}} \times X, \nu_{\frac{1}{2}, \frac{1}{2}} \times \mu)$.

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For some time, it was unclear if Kalikow constructed one example, or many examples (up to isomorphism).

It turns out that the following problem is difficult:

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In different words, the $[T, T^{-1}]$ system “remembers” the measure theoretic entropy of T . It is unclear what else it remembers.

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Are there similar results for deterministic $[T, T^{-1}]$ systems?

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In the deterministic version, one replaces the Bernoulli shift in the base by an irrational rotation.

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and the function with values 1 and -1 on each half of $\mathbb{T}$

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Definition

Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. The $[T, T^{-1}]_{R_\alpha}$ system is the skew product

$$(\theta, x) \mapsto (R_\alpha(\theta), T^{\tau(\theta)}(x))$$

acting on the product space $(\mathbb{T} \times X, \text{Leb} \times \mu)$.

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All these systems have zero entropy, regardless of the irrational α and the choice of $T \curvearrowright (X, \mu)$.

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Natural question

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- Austin proved that the $[T, T^{-1}]$ system remembers $h_\mu(T)$.
- It turns out that the $[T, T^{-1}]_{R_\alpha}$ system remembers much more: the isomorphism class of T (up to taking inverses).

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Theorem (C, soon on arXiv)

Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and let S and T be ergodic transformations. Then $[T, T^{-1}]_{R_\alpha}$ is isomorphic to $[S, S^{-1}]_{R_\alpha}$ if and only if T and S are flip isomorphic.

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Thus the deterministic version of the $[T, T^{-1}]$ construction is very different to the original one! (Bernoulli)

More generally, one can also vary the rotation in the base:

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Let $\alpha, \beta \in \mathbb{R} \setminus \mathbb{Q}$ and let S and T be ergodic transformations. Then $[T, T^{-1}]_{R_\alpha}$ is isomorphic to $[S, S^{-1}]_{R_\beta}$ if and only

- T and S are flip isomorphic.
- $\alpha = \pm\beta \pmod{1}$.

Kronecker factors

Definition

Let G be a compact Abelian group with Haar measure m_G . Consider the rotation by $\alpha \in G$

$$R_\alpha: G \rightarrow G, g \mapsto g + \alpha$$

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An ergodic measure preserving system may factor onto several Kronecker systems. Among these factors there is a biggest one, called “the” **Kronecker factor** of a system.

Fact

A factor of measure preserving systems “goes down” to their respective Kronecker factors.

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Here $(\mathbb{Z}/2\mathbb{Z}, +)$ is the group of integers mod 2, and $R_{(\alpha,1)}$ is the map

$$(\theta, n) \mapsto (\theta + \alpha, n + 1)$$

Proof ideas for the isomorphism theorem.

Let S and T be ergodic automorphisms of (X, μ) so that $[S, S^{-1}]_{R_\alpha}$ and $[T, T^{-1}]_{R_\alpha}$ are measure theoretically isomorphic.

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Prove that S and T are flip isomorphic.

Steps (informal)

- 1 Consider an arbitrary isomorphism from $[S, S^{-1}]_{R_\alpha}$ to $[T, T^{-1}]_{R_\alpha}$

$$\Upsilon: \mathbb{T} \times X \rightarrow \mathbb{T} \times X$$

Write $\Upsilon = (\phi, \psi)$.

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Write $\Upsilon = (\phi, \psi)$.

- 2 Show that ϕ is nice.
- 3 Use ψ to obtain flip isomorphisms between S and T .

Proposition (ϕ is nice)

We have

$$\phi(\theta, x) = \theta + c$$

for some $c \in \mathbb{T}$.

Recall that $\Upsilon = (\phi, \psi)$. Writing

$$\psi_{\theta}(x) = \psi(\theta, x)$$

we can see $\{\psi_{\theta} : \theta \in \mathbb{T}\}$ as a family of functions $X \rightarrow X$.

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Almost every ψ_θ is a flip isomorphism between S and T . That is,

$$\psi_\theta \circ S = T^\ell \circ \psi_\theta$$

for almost all $\theta \in \mathbb{T}$, for some $\ell \in \{-1, 1\}$.

Because Υ intertwines the n -th powers of $[S, S^{-1}]_{R_\alpha}$ and $[T, T^{-1}]_{R_\alpha}$ we have the relation

$$\psi_{\theta+n\alpha} \circ S^{\sum_{i=0}^{n-1} \tau(\theta+i\alpha)} = T^{\sum_{i=0}^{n-1} \tau(c+\theta+i\alpha)} \circ \psi_\theta$$

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Along a subsequence $(q_n)_{n \in \mathbb{N}}$ of good values, this equation becomes

$$\psi_{\theta+q_n\alpha} \circ S^1 = T^{\pm 1} \circ \psi_\theta$$

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The map $\theta \mapsto \psi_\theta$ is measurable, so one can use Lusin theorem and pretend that it is continuous.

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Proposition (Aaronson and Keane, 1982)

Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. If $(q_n)_{n \in \mathbb{N}}$ are the odd denominators in the continued fraction expansion of α , then for almost all θ , we have

$\sum_{i=0}^{q_n-1} \tau(\theta + i\alpha) = 1$ for infinitely many n 's.

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- Here we are (essentially) applying the same method to the $\mathbb{Z}^2$ -cocycle

$$\omega: \mathbb{T} \rightarrow \mathbb{Z}^2, \theta \mapsto (\tau(\theta), \tau(\theta + c))$$

and using that $(1, 1)$ or $(1, -1)$ is an essential value.

Thanks for your attention!